

Chapter 13: Spread of pathogens

Pathogen persistence

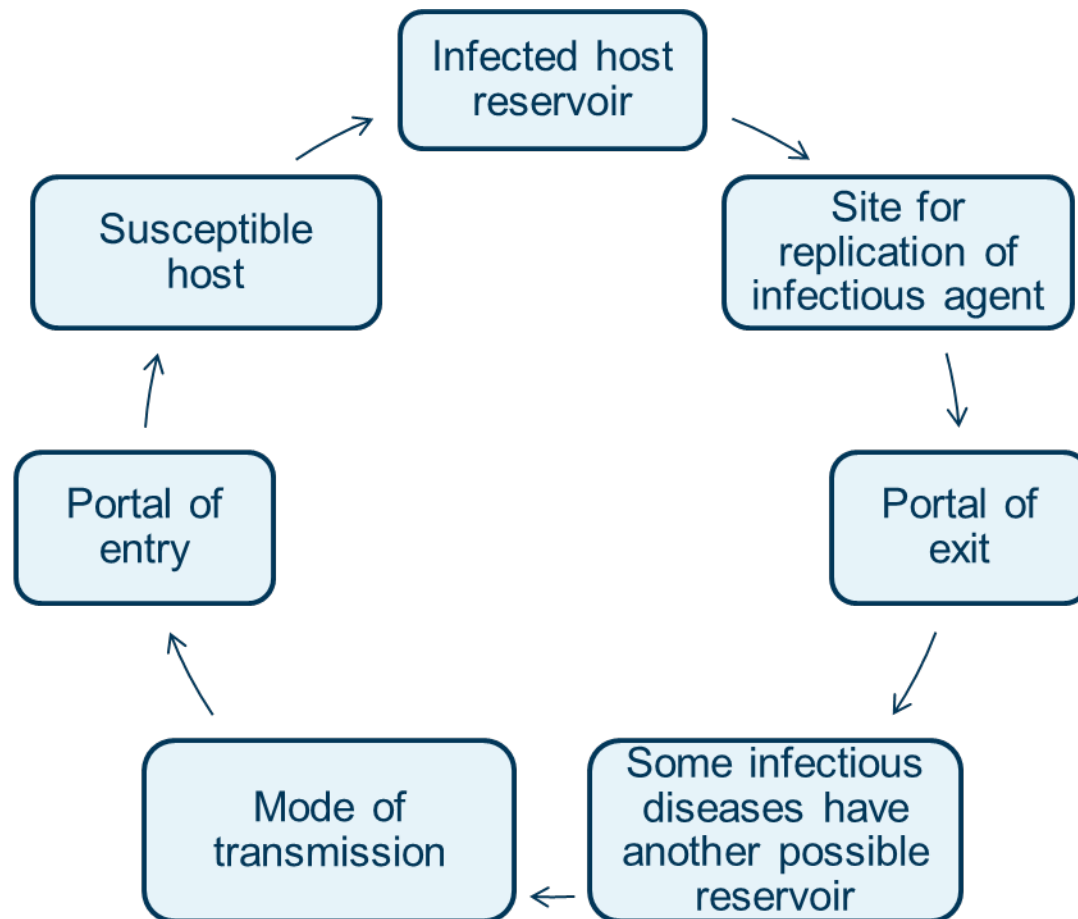
Like other organisms, pathogens strive to survive and reproduce, but pathogens usually do so at the expense of a host. The host provides an environment that assists in the survival of the pathogen. In order to best survive and reproduce, pathogens need to:

- invade (enter) the host
- exploit a nutrient-rich area of the host
- avoid host **defence mechanisms**
- replicate
- exit and transmit to new hosts.

Life cycle of a pathogen

Life cycle

This cycle is known as a pathogen's **life cycle**. Some pathogens have developed elaborate life cycles that enable them to survive, reproduce and spread.



Portals of host entry and exit

Portals of entry	Portals of exit
• Skin (bite, blood feed, penetration) • Mucous membranes • Wounds in epidermal layers of animals and plants • Eyes and ears • Respiratory system/nose/throat/lungs	• Bite/blood feed/saliva • Digestion (elimination) • Respiratory system (coughing, sneezing) • Blood contact • Reproductive system (some honeybee viruses pass from queen bee to worker bees)

Transmission

Transmission is how the pathogen is transferred from the host and/or a reservoir to a susceptible host, which can involve developmental stages occurring in the environment or in vectors.

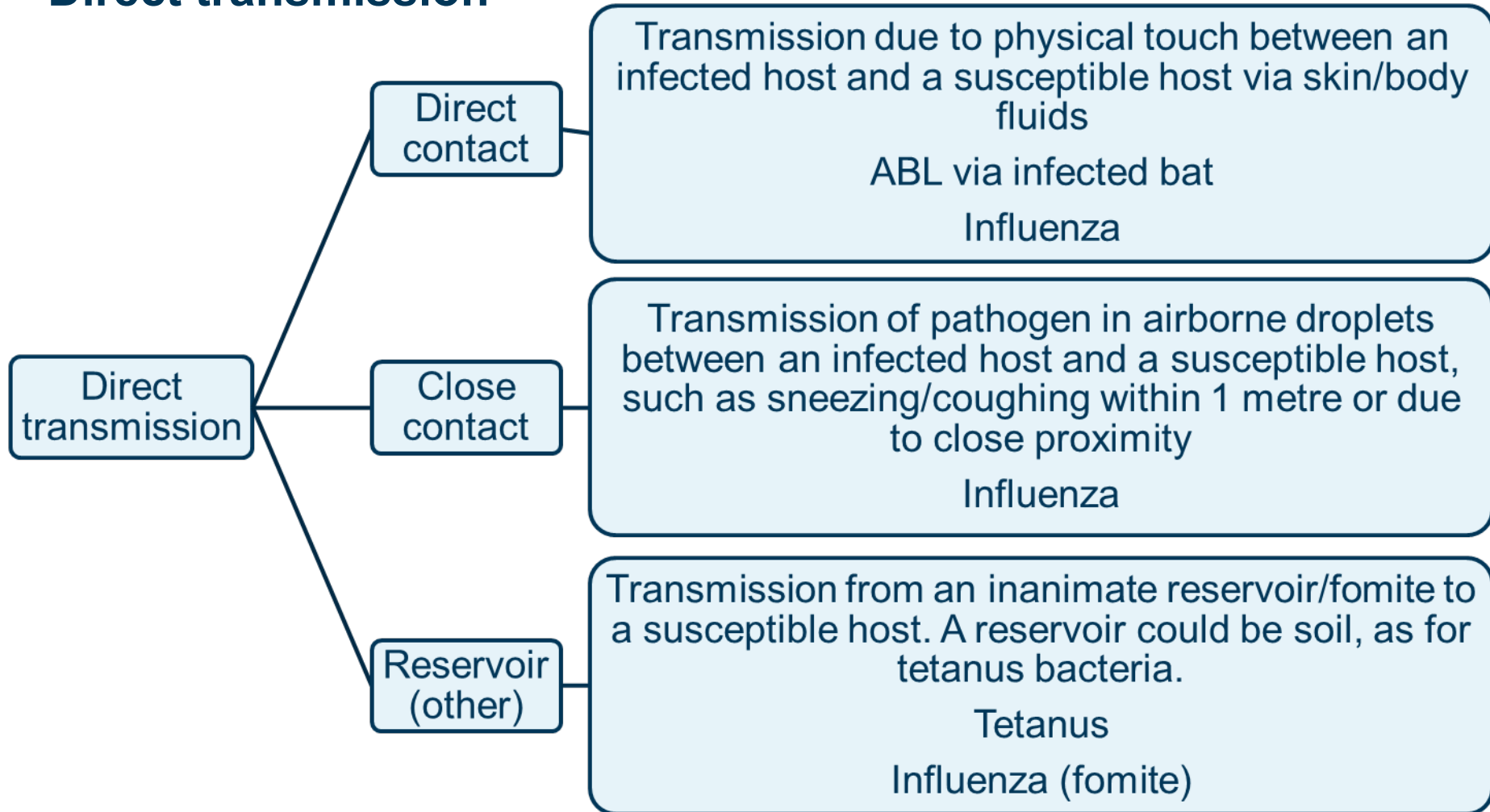
Vectors are living or non-living things that transport pathogens from the infected host to a susceptible host. Mosquito vectors transport pathogens through blood feeds. Bat vectors transport pathogens through bites and scratches.

The different modes of transmission are commonly classified as direct or indirect. **Direct** transmission is the transfer of a pathogen from an infected host, or other reservoir, to a susceptible host by direct contact or droplet spread.

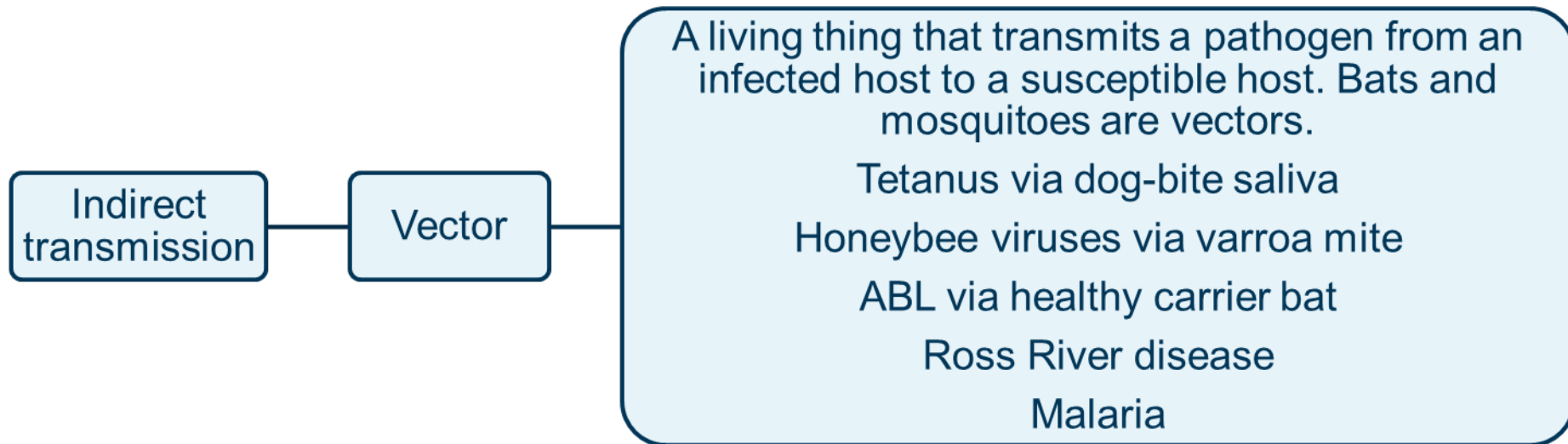
Indirect transmission is the transfer of a pathogen from a reservoir to a host through vectors (inanimate vehicles as well as living intermediaries) or suspended air particles.

Modes of transmission

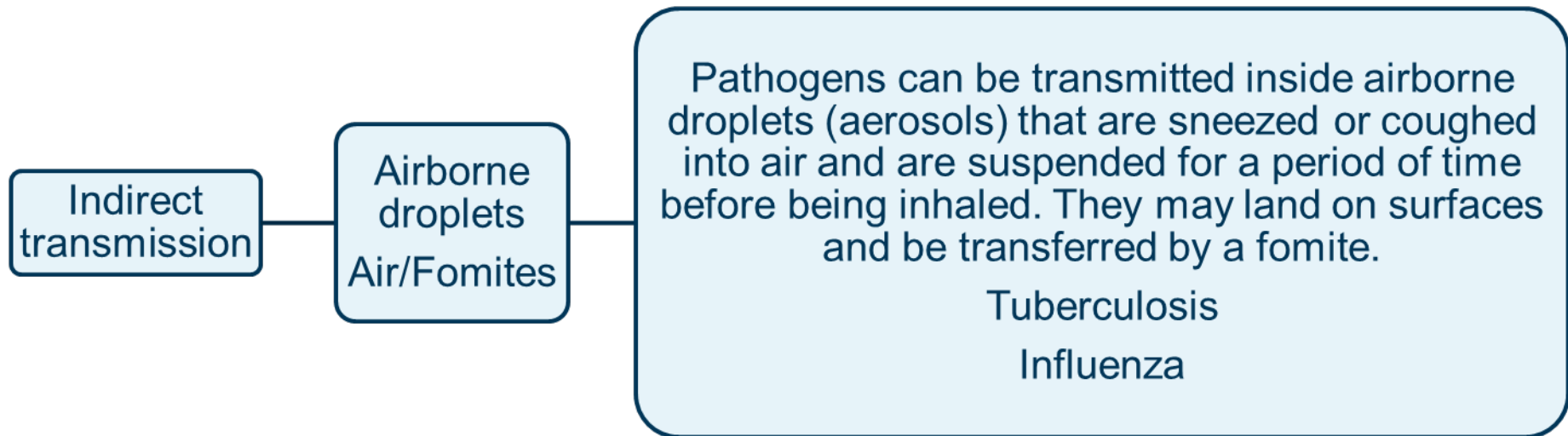
Direct transmission



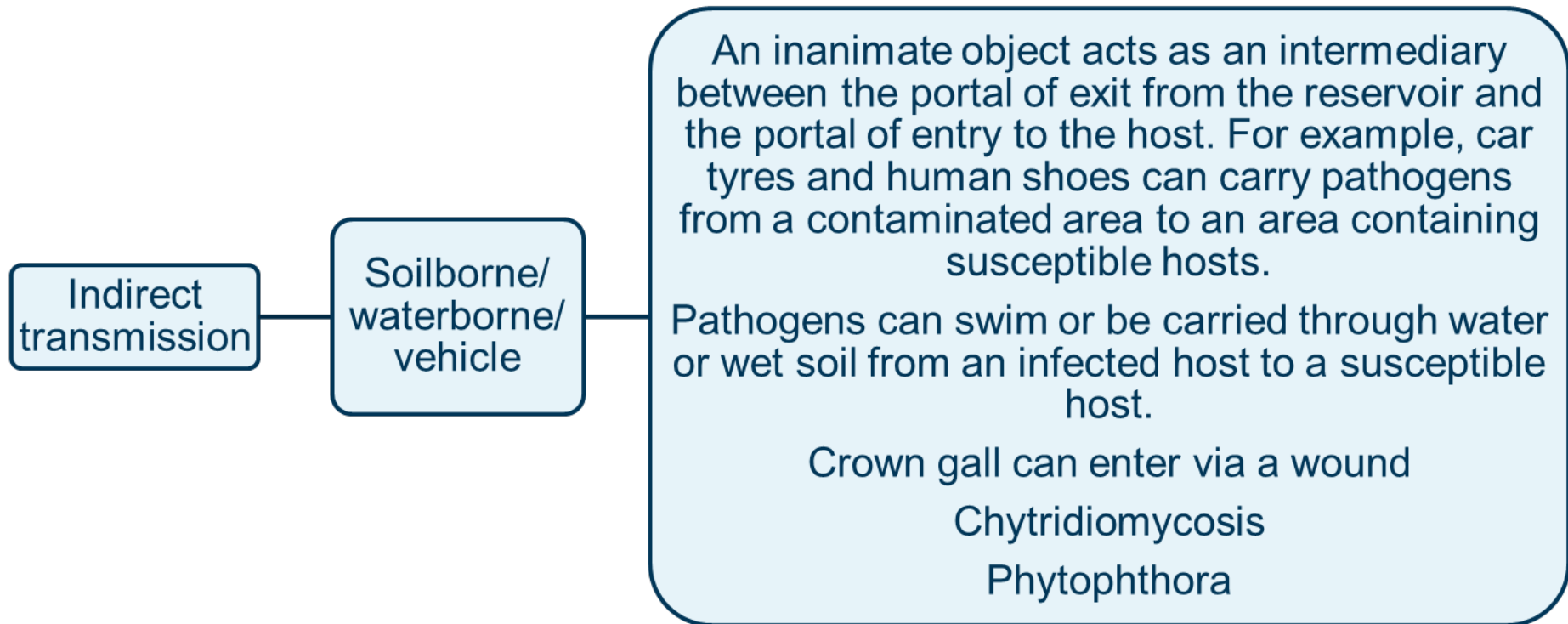
Indirect transmission: vector



Indirect transmission: airborne droplets



Indirect transmission: soilborne/waterborne/vehicle



Ten pathogen life cycles

Investigating 10 diseases

Pathogen classification	Diseases			
Virus	Influenza	Ross River disease/fever	Viral diseases of honeybees	Australian bat lyssavirus
Bacteria	Tuberculosis (TB)	Tetanus	Crown gall of plants	
Fungus	Chytridiomycosis (amphibian chytrid fungus disease) (caused by fungi)			
Protist	Malaria	Phytophthora dieback (jarrah dieback)*		

*The Phylum Oomycota containing Phytophthora dieback has been removed from the Fungi Kingdom and placed in the Protista Kingdom

1 Influenza

The infectious agents that cause influenza are the influenza A, B or C viruses.

The single-stranded RNA virus is usually transmitted from a respiratory tract, through the air inside airborne droplets, when an infected host coughs or sneezes. The viruses, found in the airborne droplets/spray, are inhaled.

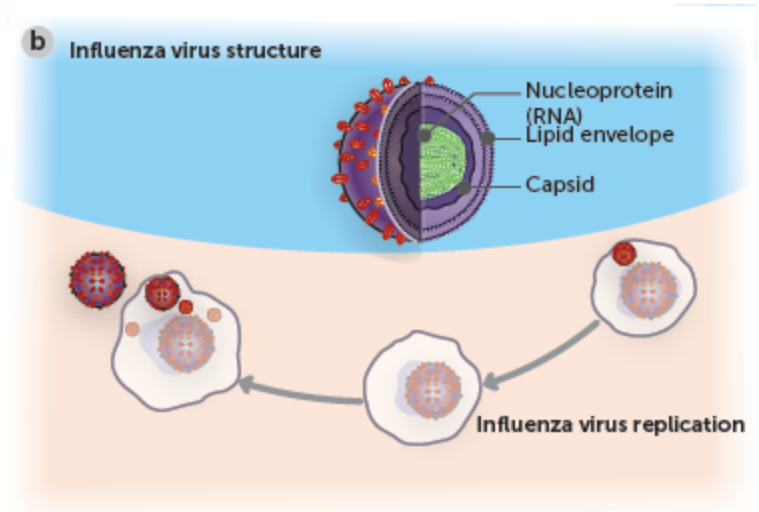
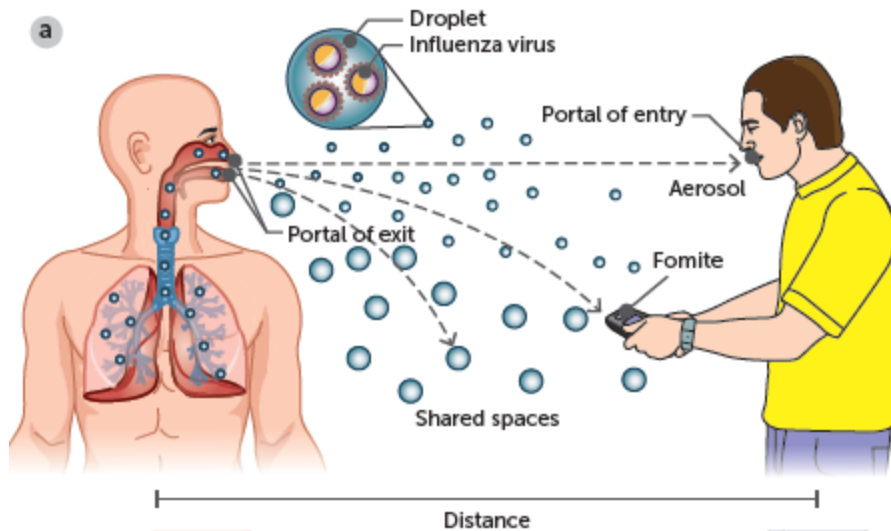
The viruses will attach to epithelial cells along the respiratory tract if those cells contain suitable virus receptor cells. The influenza virus is an RNA-type virus. It will undergo viral replication inside the epithelial cells.

After the release of new viruses from the epithelial cells, a cough or sneeze may send a spray of airborne droplets into the air. The droplets may be inhaled by someone in close proximity or become an aerosol and drift with air currents, to be inhaled at a later time or land on a surface known as a fomite.

Therefore, transmission, although usually direct via close contact, can be indirect via airborne droplets or fomites.

Ten pathogen life cycles

Influenza life cycle



2 Ross River disease

The infectious agent that causes Ross River disease (also known as Ross River fever) is Ross River virus. It is a single-stranded RNA virus.

The virus is an alphavirus and is endemic in Australia, always present at low levels in some populations or regions.

The virus is transmitted by female mosquitoes, usually from marsupials such as wallabies or kangaroos, to humans via blood feeds.

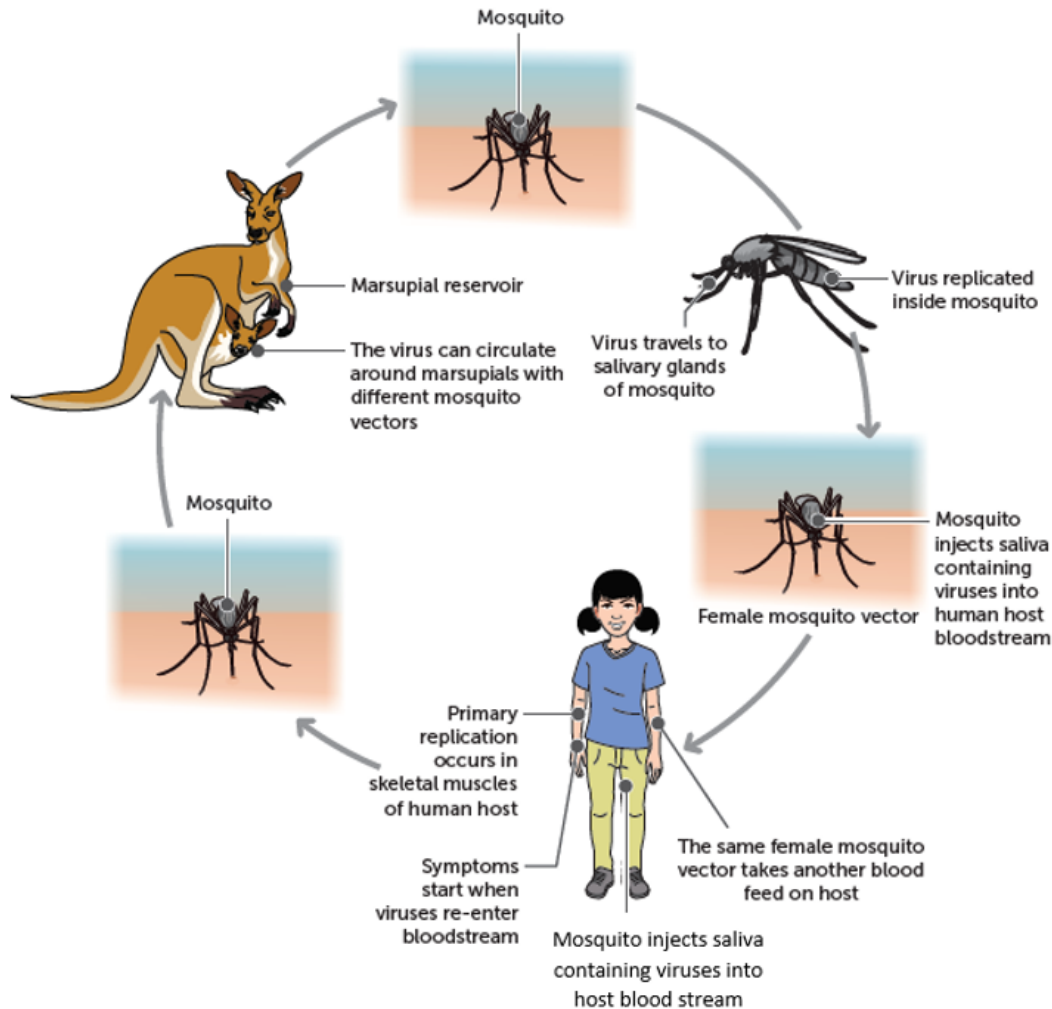
Outbreaks in Australia have coincided with an increase in rainfall and flooding.

After significant rainfall, the increase in still water bodies increases the potential breeding sites for the mosquito vectors.

The disease is endemic in the south-west of WA, where outbreaks have been recorded, such as the one in 2017.

Ten pathogen life cycles

Ross River virus life cycle



3 Viral diseases of honeybees

There are many infectious agents that cause honeybee diseases, including 24 known honeybee viruses.

Common diseases are sacbrood, black queen bee virus, chronic bee paralysis virus and deformed wing virus.

Viruses may cause diseases that are asymptomatic, or they can be highly virulent and kill the hive.

This significantly reduces the rate of pollination in the areas where the bees had previously resided, including the pollination of crops, which in turn affects human populations.

Sacbrood disease

Sacbrood disease is caused by the sacbrood virus, which has infected bees internationally and was detected in WA in 1979.

This disease mostly affects worker larvae, but can also infect adult honeybees.

Infected larvae die just before pupation. The virus's mode of transmission may be foodborne: the virus can be carried by vector nurse bees when contaminated food is fed to the brood of larvae.

The virus replicates inside the larvae cells, causing the larvae to display unusual behaviour.

The larvae turn onto their backs and lie stretched out with their heads lifted.

After the larvae die and dry out, millions of viruses surround the dead larvae. When humans clean out dead larvae, they can cause the viruses to be disturbed and transported to other worker bees.

Ten pathogen life cycles

Sacbrood disease



Rob Snyder/Bee Informed Partnership, University of Maryland (www.beeinformed.org) accessed May 2020.

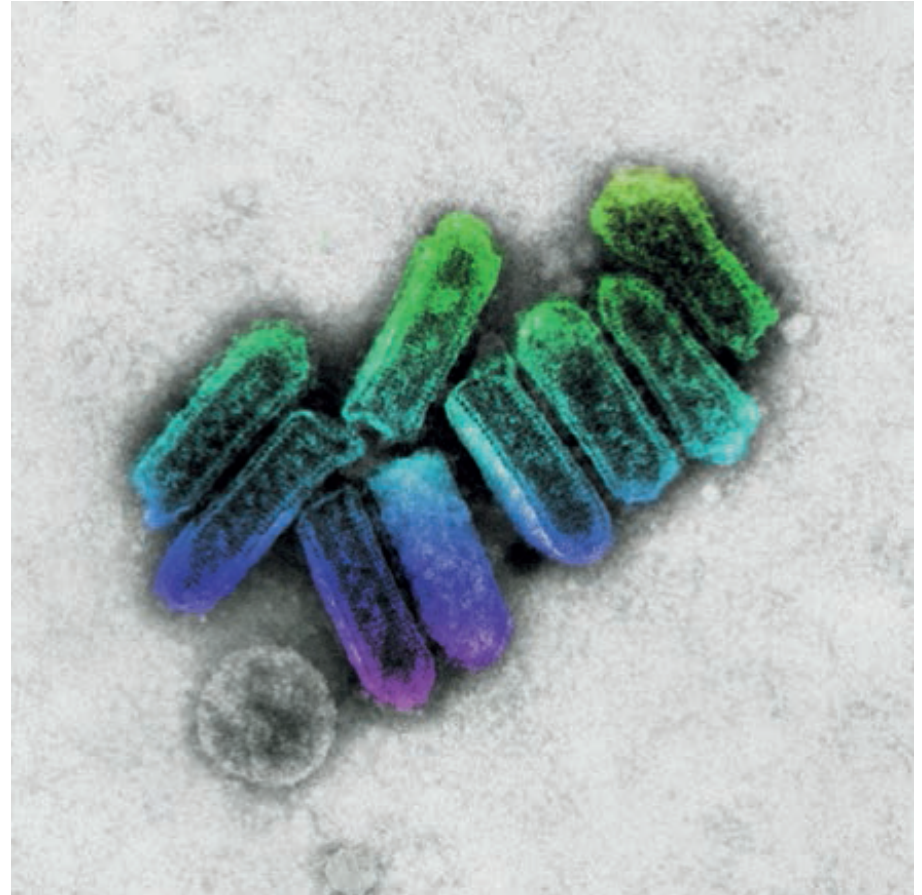
4 Australian bat lyssavirus (ABL) disease

Australian bat lyssavirus (ABL) disease is caused by the Australian bat lyssavirus.

The virus was discovered in 1966. It is closely related to the rabies virus, but the rabies virus is not found in Australia.

The virus infects Australian flying foxes (fruit bats) and microbats.

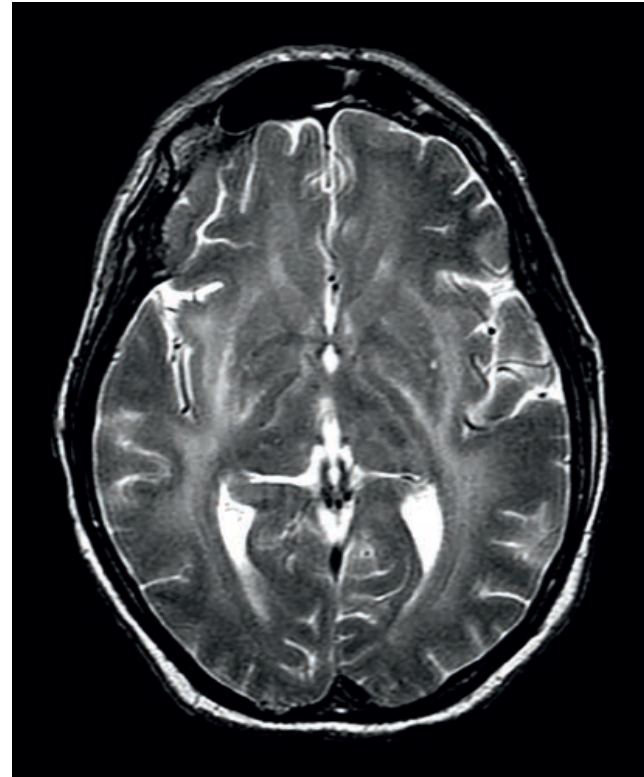
The disease can be transmitted from bats to humans, horses and other bats.



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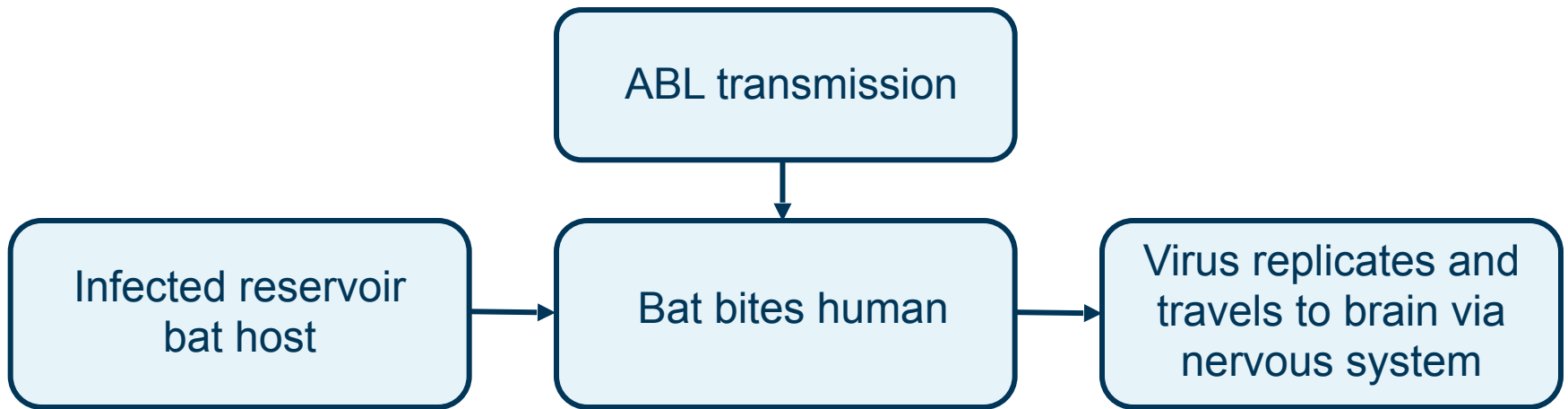
ABL disease in humans

The disease affects the nervous system in humans, and can cause fatal encephalitis (inflammation of the brain), shown at right below.



Science Photo Library/Living Art Enterprises, LLC/Science Source

Australian bat lyssavirus life cycle



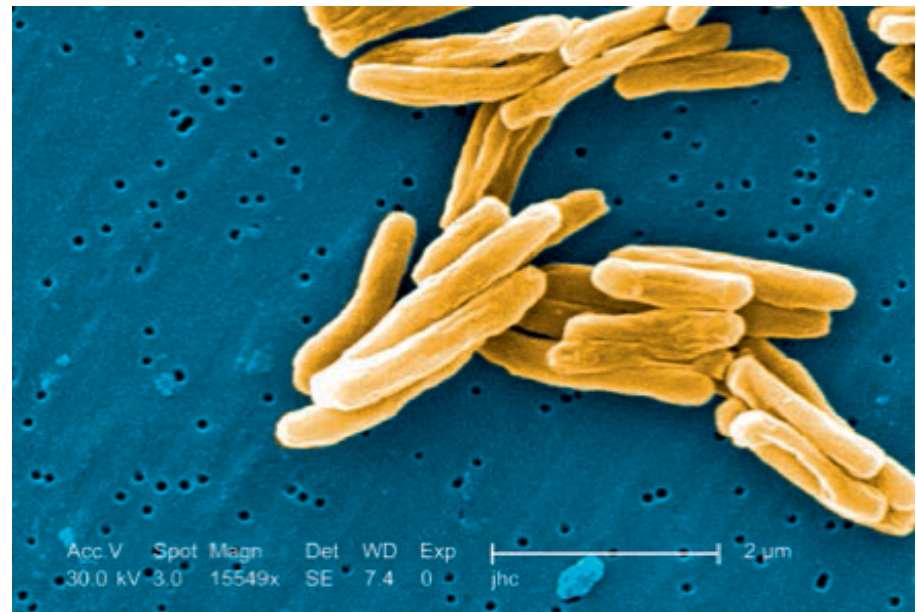
Ten pathogen life cycles

5 Tuberculosis (TB)

The disease tuberculosis (TB) is caused by the infectious agent *Mycobacterium tuberculosis*. It is not common in Australia, but is very common in many developing countries.

The tubercle-producing, rod-shaped bacterium mainly affects the lungs of the respiratory system.

(A tubercle is a small round structure made of cells that is produced as a result of the infection.)



Alamy Stock Photo/Scott Camazine

Tuberculosis

When a person with tuberculosis coughs, tubercle bacilli are transported in fine droplets from the infected human host's lungs into the air, from where they can be inhaled by a susceptible person.

Direct transmission occurs when a person is standing only 1–2 metres away from the infected person and inhales the airborne droplets.

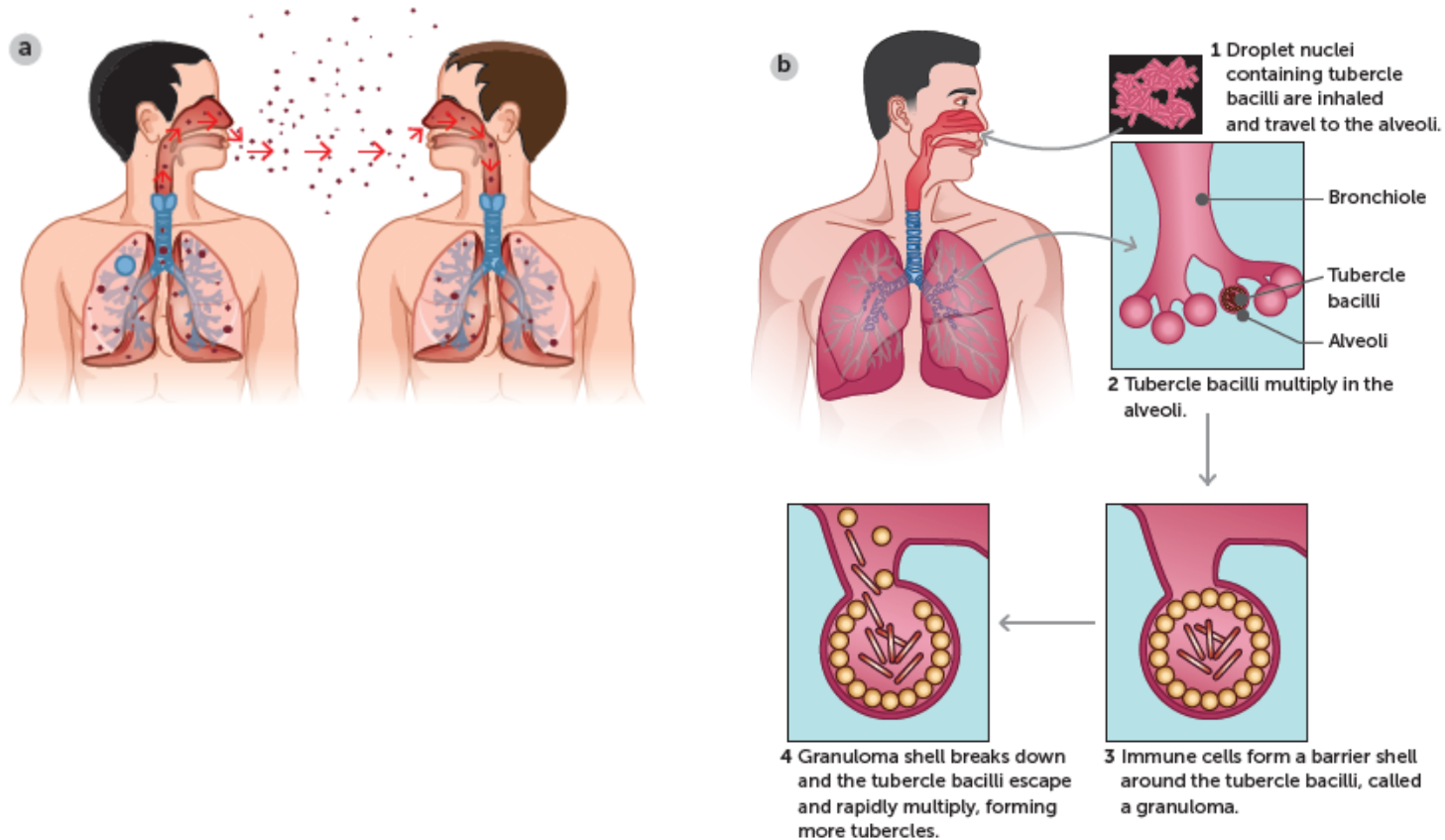
The TB bacteria rapidly move through that person's mouth, throat, trachea and bronchi and deep into the lungs.

Indirect transmission can also occur when an infected person coughs, sending droplets into the air, because the bacilli may land on a dusty surface.

When the dust is disturbed and flies in the wind, it can carry the bacteria into the air and be inhaled.

Ten pathogen life cycles

Mycobacterium tuberculosis life cycle



6 Tetanus

The disease tetanus is caused by the infectious agent *Clostridium tetani*.

Clostridium tetani is an obligate anaerobic Gram-positive bacillus.

Entry into a susceptible host involves contact with the spore form of the bacterium, universally present in soil, typically via a deep wound.

The spores are also found in animal intestines and faeces.

The bacterial spores will not germinate in a superficial wound because they are anaerobic bacteria and will only germinate and multiply in hypoxic (low oxygen) conditions.

Wounds that penetrate deeply into skin and soft tissues provide the hypoxic conditions required for germination of the spore and a portal of entry.

Ten pathogen life cycles

Clostridium tetani life cycle



1 Soil reservoir

Soil particles containing bacteria spores can be found on rusty nails and on dog's teeth.



2 Mode of transmission

Entry portal is via a deep wound in the skin involving direct contact with the soil reservoir (such as by stepping on a nail or being bitten by a dog).

Ten pathogen life cycles

Clostridium tetani life cycle



3 Reproduction

Inside the deep, hypoxic wound, bacteria spores germinate, grow and multiply asexually via binary fission.



4 Production of toxin

During the growth of the bacteria, neurotoxins are produced (e.g. tetanospasmin), which travel inside neurons from the peripheral nervous system (PNS) to inhibitory interneurons in the central nervous system (CNS).

Ten pathogen life cycles

***Clostridium tetani* life cycle**



5 Impact on host

The toxin blocks inhibitory neurotransmitters and nerves fire continuously, producing rigidity, unopposed muscle contraction and spasm, including 'lockjaw' (muscles contract and can't relax – now called trismus). This can affect respiration/breathing and lead to death.

6 Infectious, not contagious

The pathogen does not normally transmit from human to human.

7 Crown gall disease

Crown gall disease of plants is caused by the infectious agent *Agrobacterium tumefaciens*.

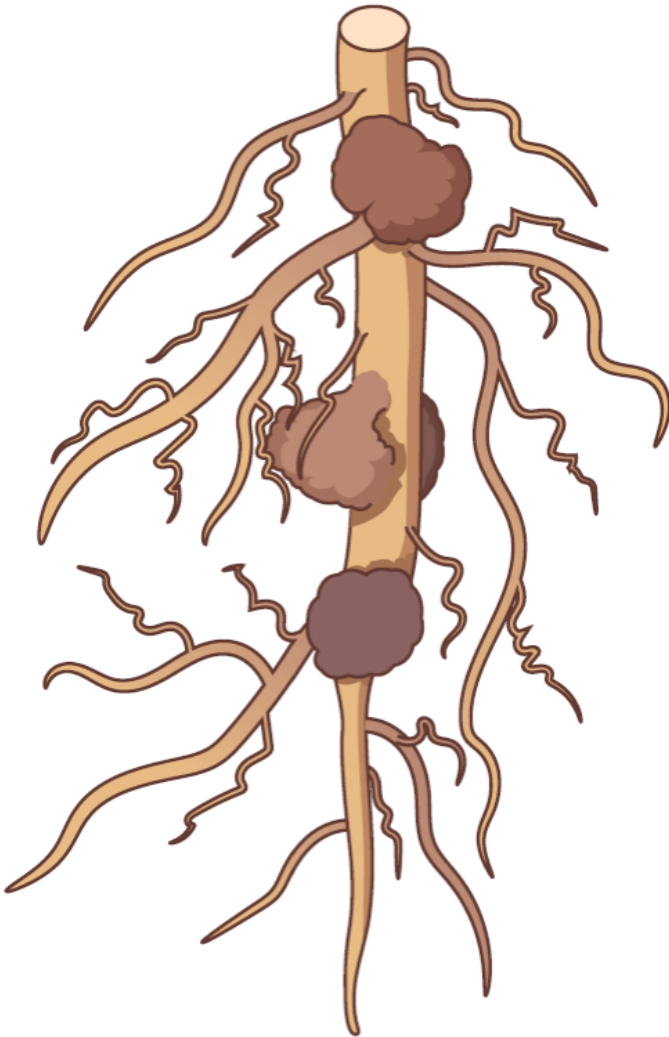
Bacteria can be transmitted from one host to another directly (via infected plant root to susceptible plant root contact) or indirectly (via grafting tools, car tyres, infested soil reservoirs, or movement of infested soil [such as by work boots or rain]).

Like the tetanus bacterium, *A. tumefaciens* requires a wound as a portal of entry. Unlike tetanus, the wound does not have to be deep.

The disease can spread through susceptible crops such as stone fruits and roses.

The disease can be found throughout Australia and has been a minor disease in roses and stone fruits in WA.

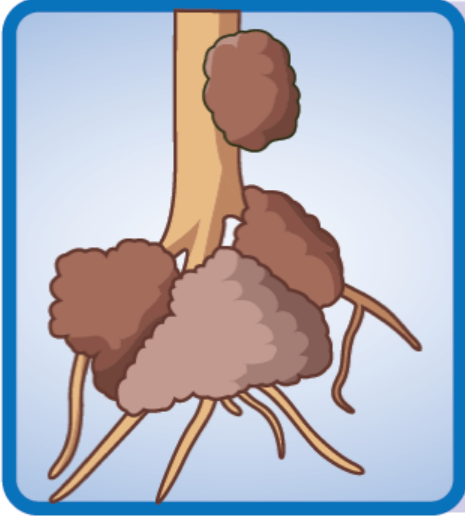
Crown gall disease



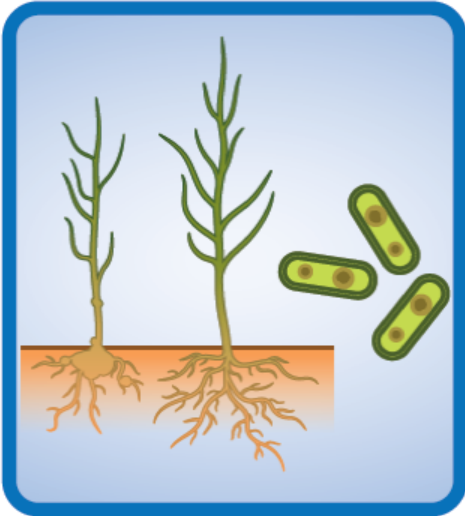
This illustration shows severe gall infection at the crown area on the roots (the point at the soil line where the main roots join the stem).

Ten pathogen life cycles

Agrobacterium tumefaciens life cycle



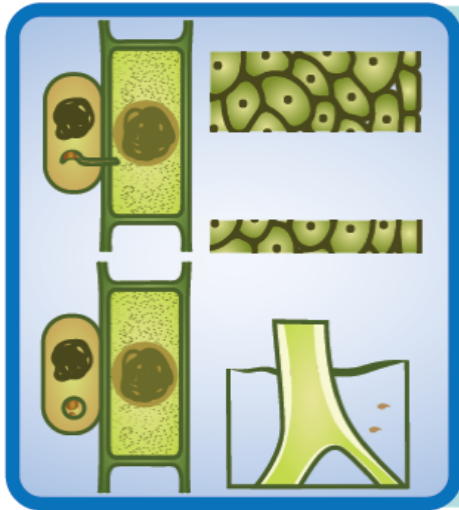
1 Bacterial spores are released from infected plant galls.



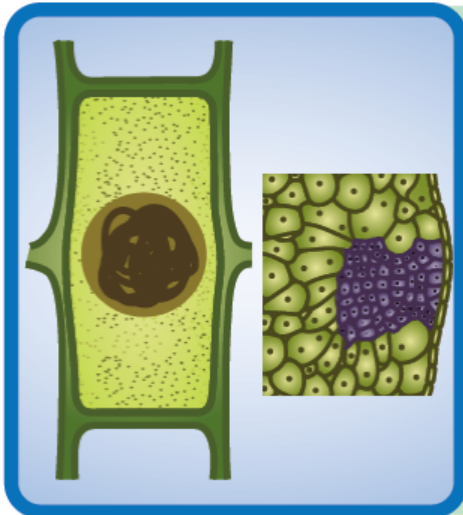
2 Direct/indirect modes of transmission – spores are transmitted by contact or by vehicle into wounded roots in a susceptible plant – the portal of entry through a wound.

Ten pathogen life cycles

Agrobacterium tumefaciens life cycle



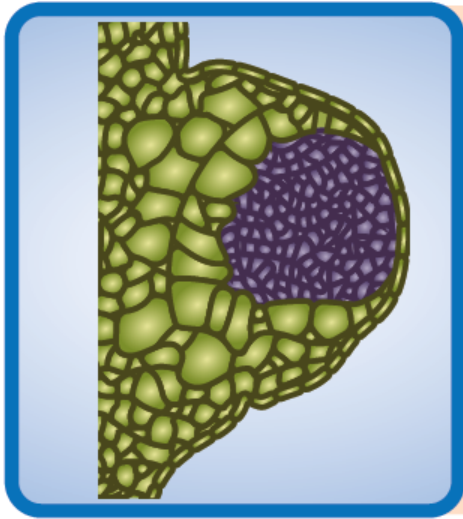
3 Bacterium attaches to a wounded plant cell and multiplies.



4 Bacterium transfers a plasmid into plant cells. Plasmid contains genes for uncontrolled cell growth.

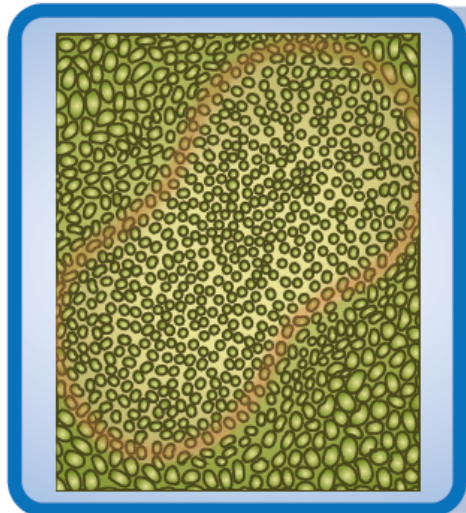
Ten pathogen life cycles

Agrobacterium tumefaciens life cycle



5 Plant cell genome transforms and starts to divide, rapidly forming galls.

6 Galls grow around the crown of the plant and stunted plant growth results.



7 Spores can be released to continue the cycle.



8 Chytridiomycosis

The disease chytridiomycosis is caused by the fungus *Batrachochytrium dendrobatidis*.

The chytrid fungus typically lives in water or soil and usually reproduces asexually.

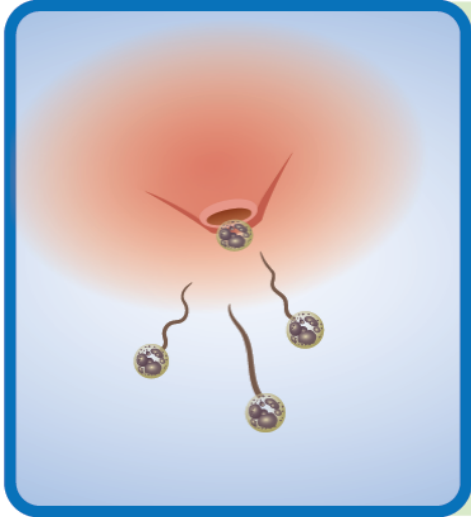
Its zoospores have a single posterior flagellum, a whip-like tail, which provides the zoospore with locomotion (makes it motile and able to swim through the water).

The disease has been detected in all states of Australia and is affecting amphibians worldwide. In WA, the main area affected is the cooler south-west, possibly because the fungus is temperature sensitive.

The disease causes amphibians to die, and at times mortality in a population is 100%.

Ten pathogen life cycles

Chytridiomycosis life cycle



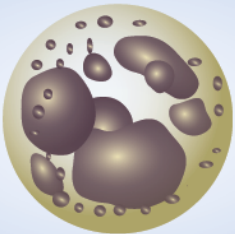
1 Zoospores are produced and released from a mature zoosporangium (thallus) discharge tube.



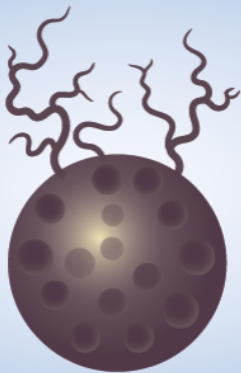
2 Zoospores swim a short distance or are carried by water currents.

Ten pathogen life cycles

Batrachochytrium dendrobatidis life cycle



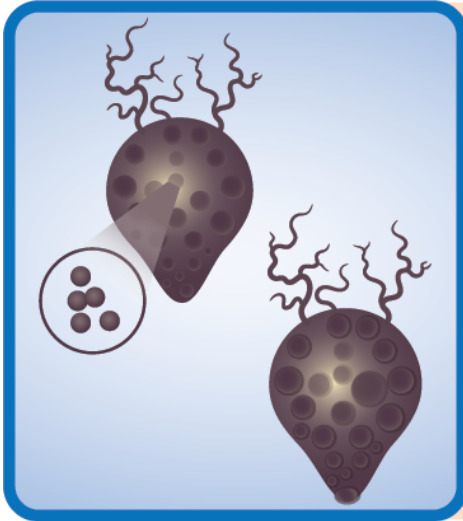
- 3** Zoospores encounter a new susceptible host and attach to and penetrate a skin cell. This is the portal of entry.



- 4** The cells invade the skin, absorbing nutrients. A new thallus develops via asexual reproduction.

Ten pathogen life cycles

Batrachochytrium dendrobatidis life cycle



5 The thallus matures and zoospores are produced.



6 Chytridiomycosis damages the frog's skin causing reduced respiration and osmoregulation, leading to death.

9 Malaria

The disease malaria is caused by unicellular protists from the *Plasmodium* genus.

Plasmodium species belong to a Protistan group resembling animal cells. The animal-type protists can be referred to as protozoa.

The infectious agents are transmitted indirectly to a host by the bite of a female *Anopheles* mosquito vector.

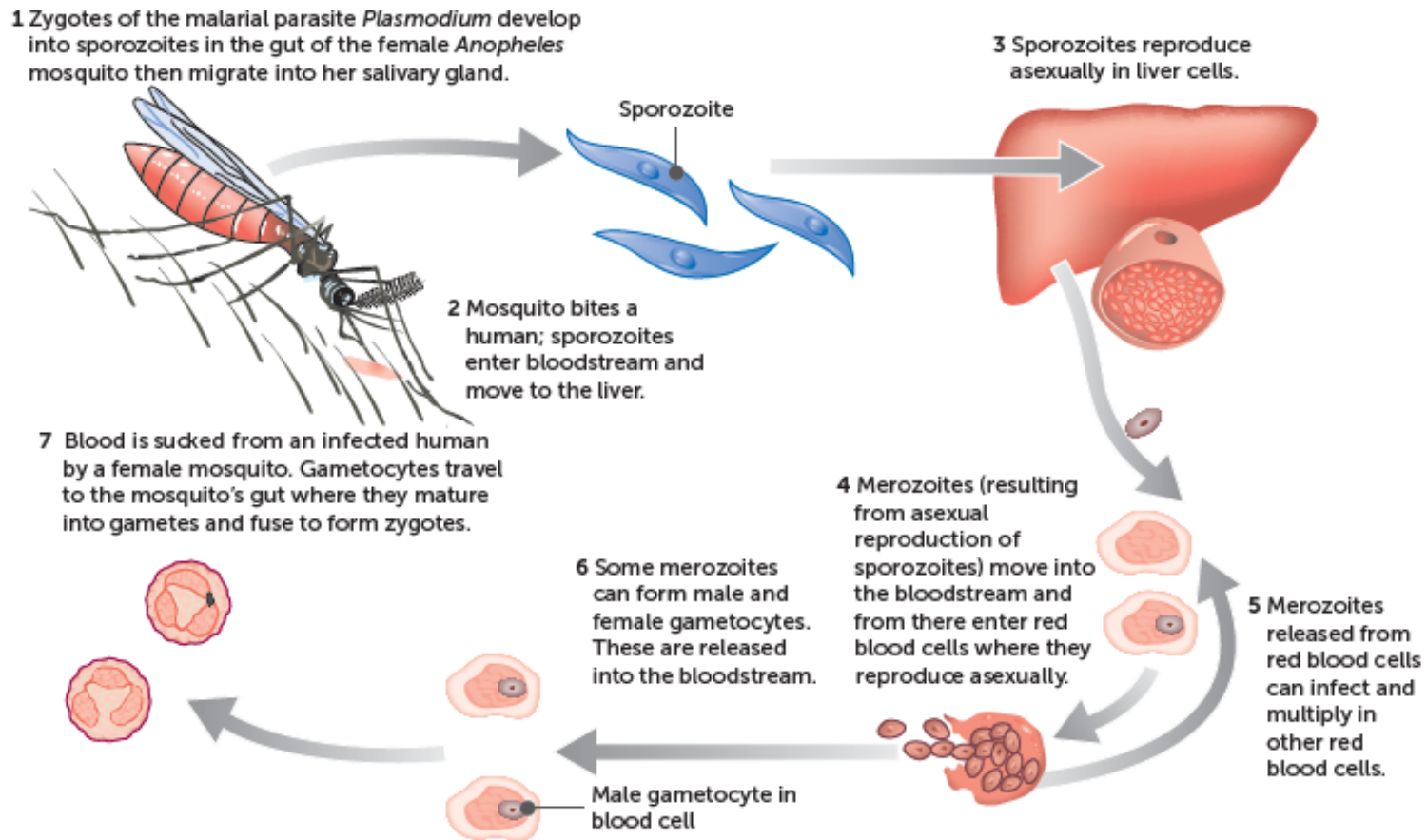
The parasite requires both the mosquito and the human host. Sexual reproduction occurs in the female mosquito gut, which is why the mosquito is known as the definitive host.

Asexual reproduction takes place in the liver and red blood cells of the human host, who is known as the intermediate host.

Malaria kills one child every 2 minutes globally. WHO has a target to eliminate malaria by 2030.

Ten pathogen life cycles

Plasmodia life cycle



Ten pathogen life cycles

10 Phytophthora dieback (jarrah dieback)

Phytophthora (pronounced fy-TOFF-thora) dieback is a disease caused by the water mould protist *Phytophthora cinnamomi*.

It is a plant pathogen that affects plants in Australia such as jarrah, banksia and grass trees, by attacking their root systems. In WA, more than 40 per cent of native plants are susceptible to the disease, particularly in the state's south-west.

Areas currently experiencing spread are along the coast in Two People's Bay Nature Reserve and in metropolitan Perth bushland areas such as Lightning Swamp Bushland.

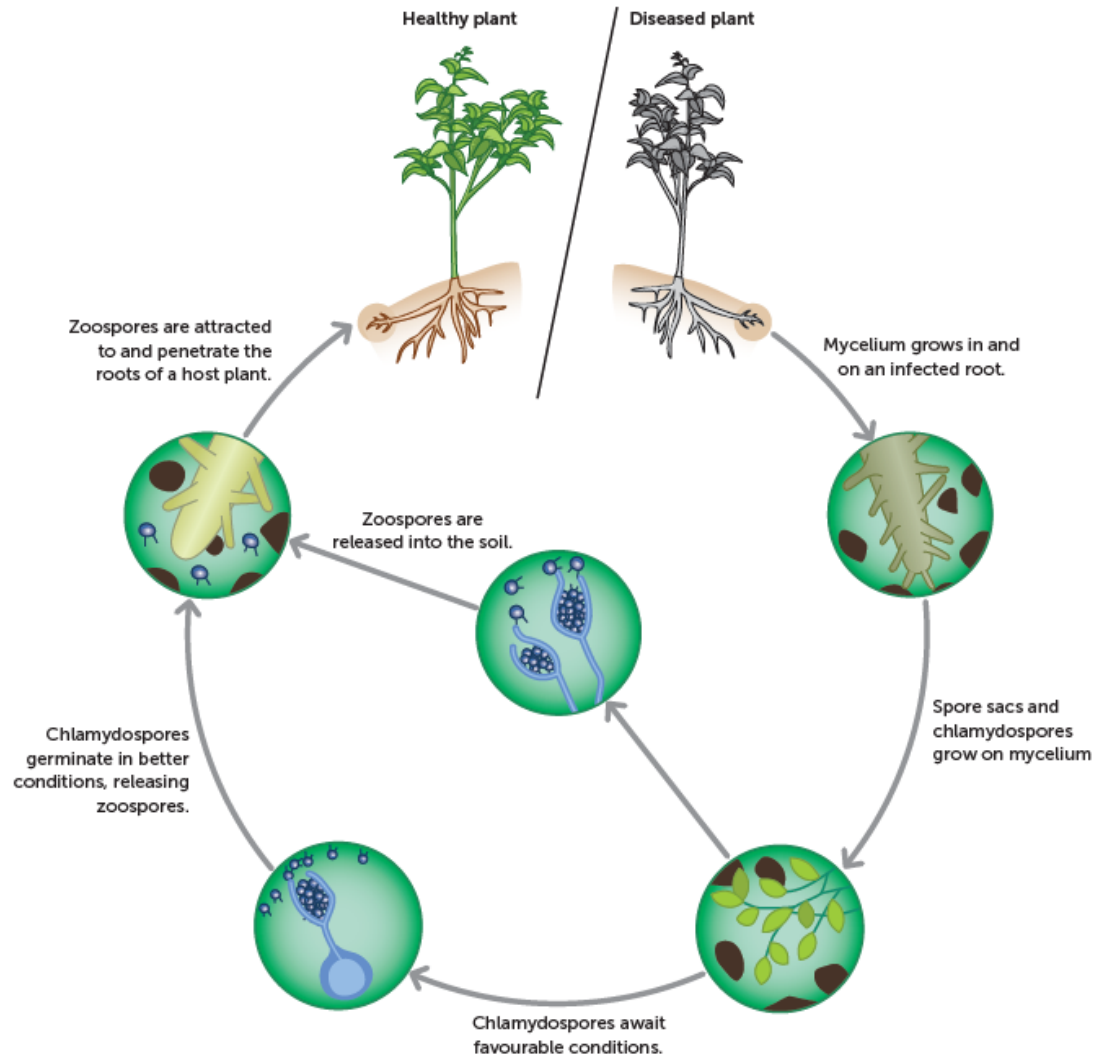
After infecting a plant, it grows rapidly in the roots and causes rapid death.



Elizabeth Bush, Virginia Polytechnic Institute and State University, Bugwood.org. Licensed under a Creative Commons Attribution 3.0 License. (<https://creativecommons.org/licenses/by/3.0/>)

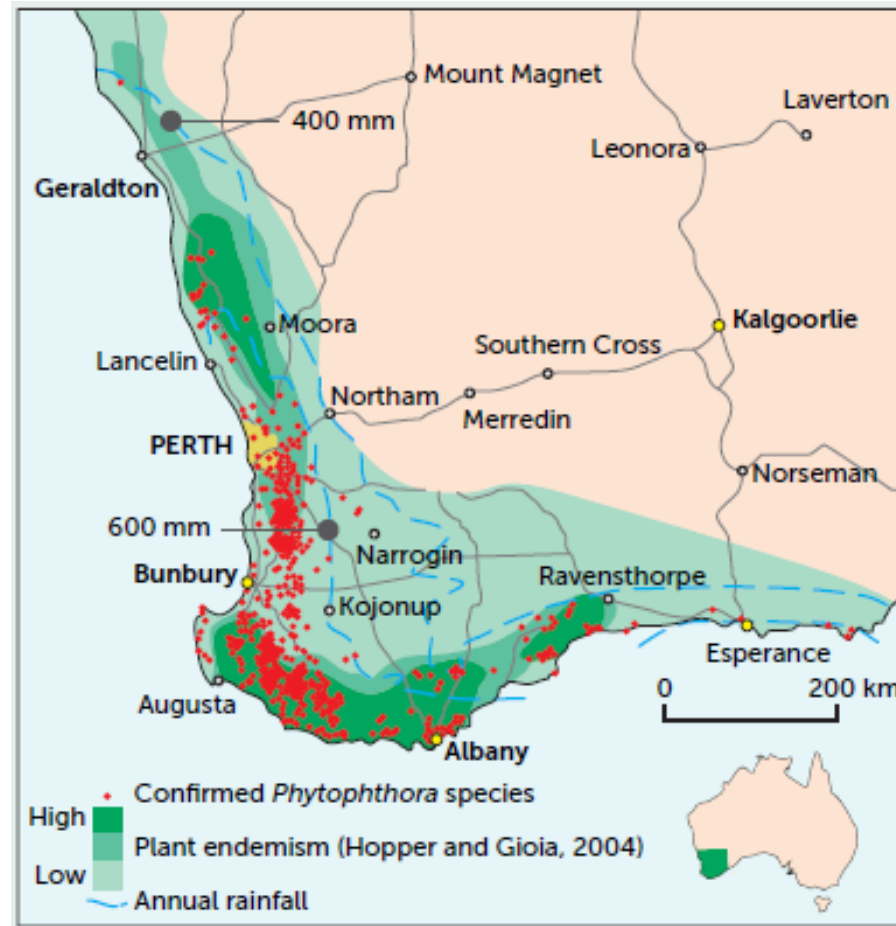
Ten pathogen life cycles

Phytophthora life cycle



Ten pathogen life cycles

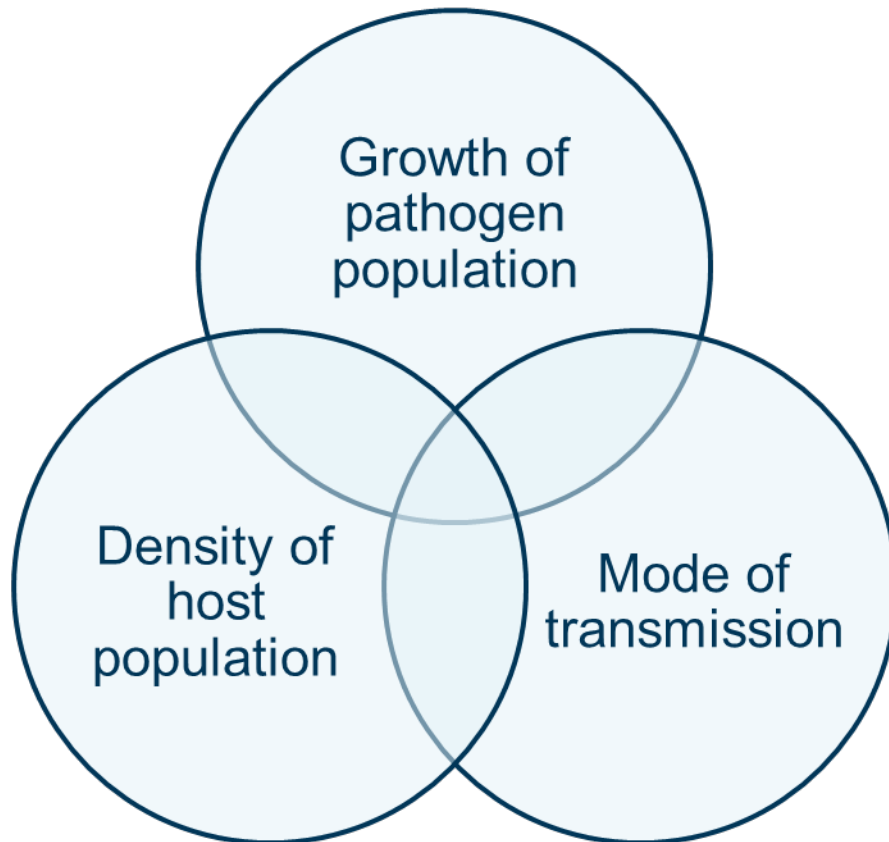
Map of *Phytophthora* distribution in WA



Factors that affect the spread of disease

Interrelated factors

The rate of spread of a specific disease is affected by a range of interrelated factors, including growth of the pathogen population, density of the host population and modes of transmission.



Interrelated factors are those that depend on and have an effect on one another.

All play a vital part in spread. They need to occur together for spread to be possible.

The effect of globalisation

Globalisation is the process by which the world is becoming increasingly interconnected as a result of massively increased trade, economic, travel and cultural exchange.

The efficiency, speed and reach of modern transport puts people at risk from the emergence of new strains of familiar diseases, or from completely new diseases. In addition, the global growth of economic activity, tourism and human migration is leading to ever more cases of the movement of both disease vectors and the diseases they carry.

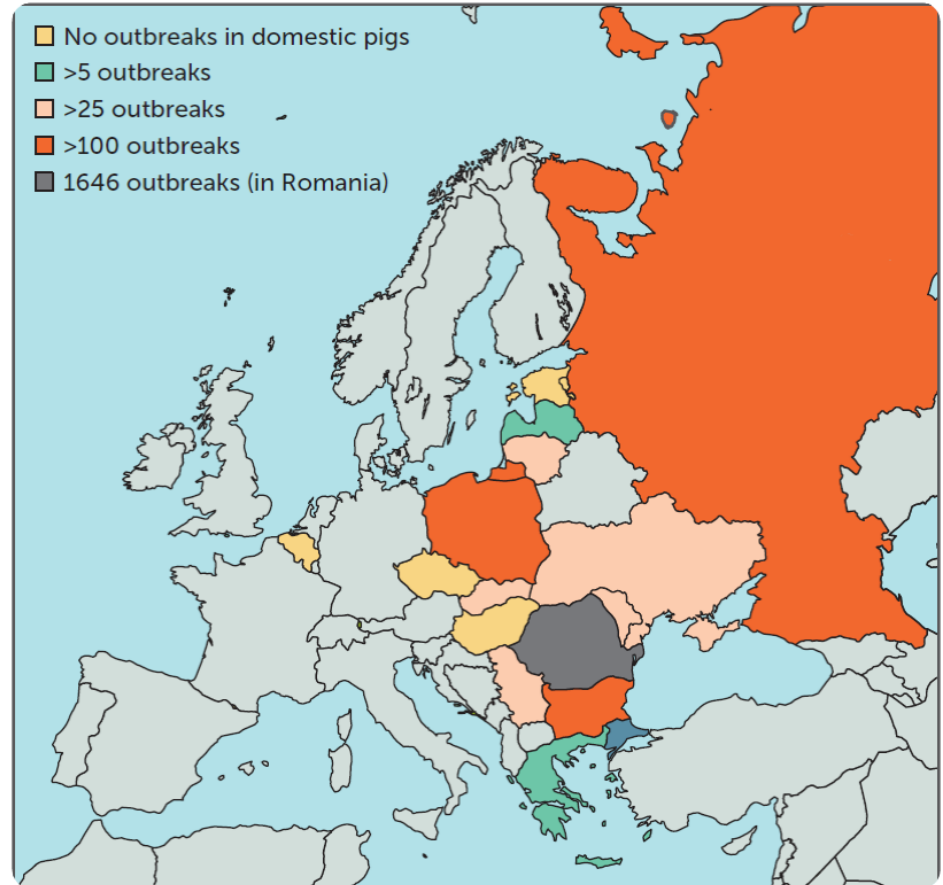
Factors that affect the spread of disease

Effect of globalisation example:

African swine fever (ASF) virus is endemic to sub-Saharan Africa. It can spread directly through live or dead pigs and their body fluids. It can also be spread indirectly through animal feed that contains pork infected with ASF, or via fomites such as transport storage surfaces, shoes and clothes. The virus can survive in extreme, dry temperatures.

The recent spread of ASF beyond Africa has been the biggest animal disease outbreak in history. It has

killed more than a quarter of the world's pigs. This has occurred due to the movement of pigs and pig products across borders.



Factors that affect the spread of disease

The effect of global climatic changes on distribution of mosquito-borne diseases

The distribution of mosquito-borne diseases, such as malaria and Ross River disease/fever, are affected by global climatic changes.

Malaria is transferred by the female *Anopheles* mosquito vector.

Mosquitoes are more active and restricted to a warm environment and therefore are sensitive to climate change.

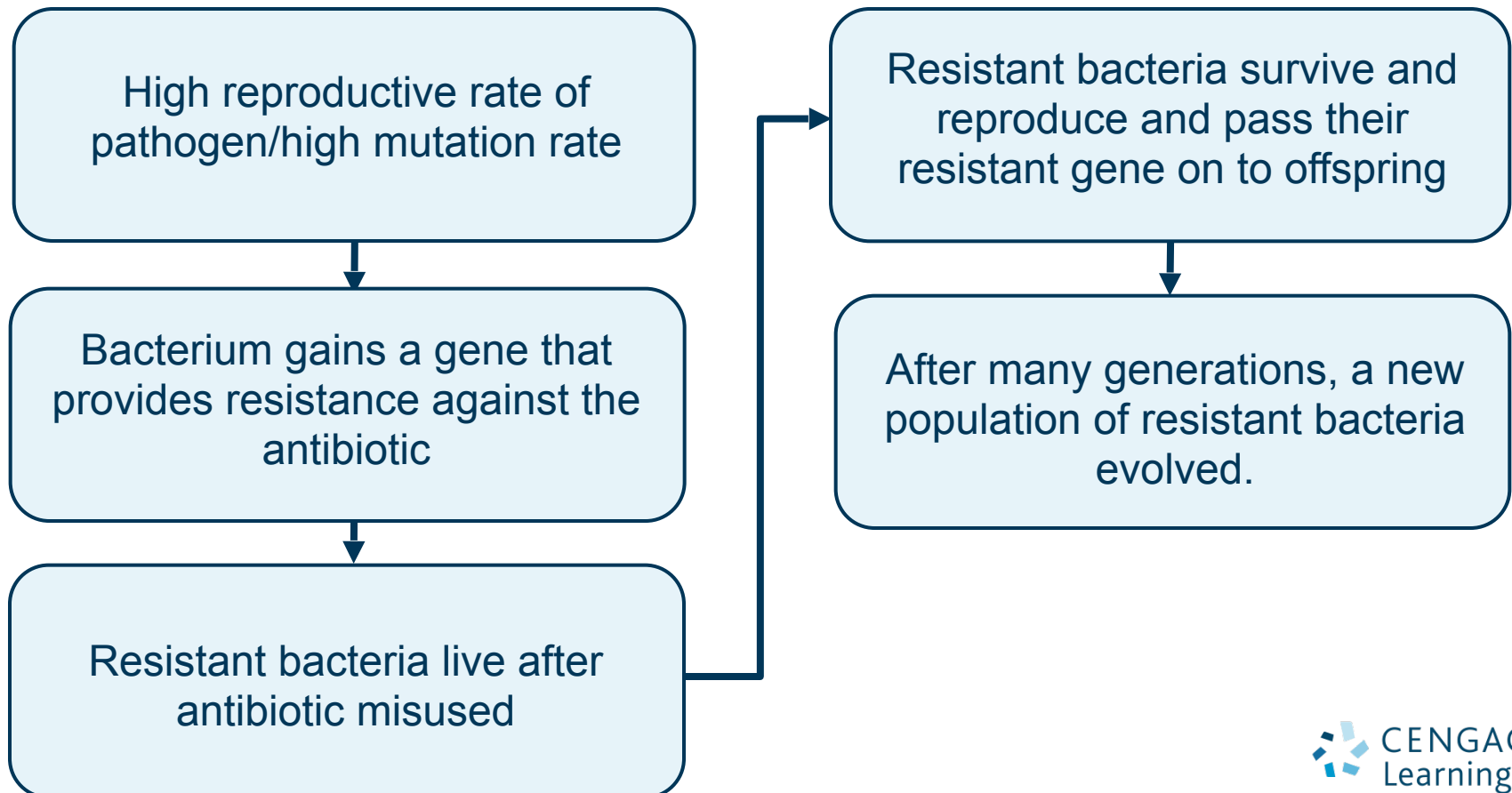
Global climate change will increase the average global air temperature, including higher at altitudes.

Mosquitoes will spread to higher altitudes as they get warmer, because mosquitoes will breed in new areas if they are warm enough.

Factors that affect the spread of disease

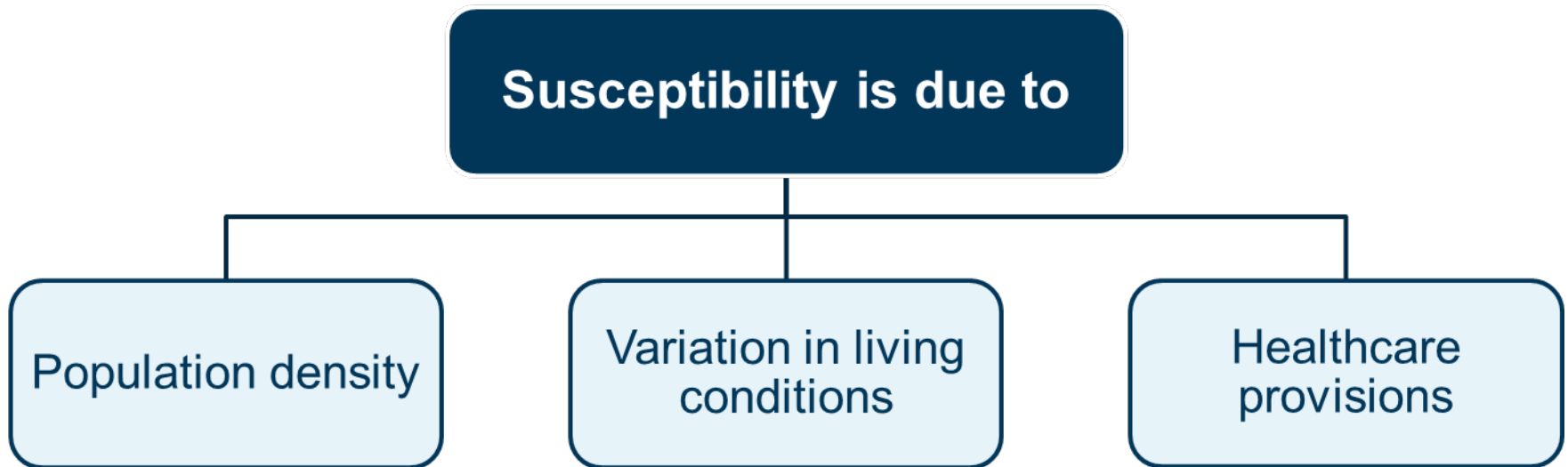
Many pathogens evolve rapidly in a changing environment

Antibiotics have been misused over time, and this has led to the evolution of bacteria and their resistance to antibiotics.



Host susceptibility is on the increase

Susceptibility of urban areas to epidemics and pandemics of infectious disease can be due to population density, variation in living conditions and healthcare provisions.



Factors that affect the spread of disease

Population density

With a higher density of humans living together, this leads to a greater contact rate between infected and uninfected individuals, which leads to the opportunity for infections to spread faster.

A higher density means an increase in interactions between animals, humans, and ecosystems.

An outbreak in a rural setting may be controlled quickly and easily, compared with an outbreak in an urban setting, simply due to the absence of susceptible hosts, particularly if the mode of transmission is direct via contact or close contact.

Factors that affect the spread of disease

Variation in living conditions

Poor living conditions can lead to the spread of infectious disease, especially airborne diseases such as TB, and vector-borne diseases such as malaria.

In developing countries, there is less access to clean, treated water. Sanitation facilities are inadequate.

Consequently, urban slums usually have high levels of faecal matter in the water consumed by households. Waterborne diseases spread rapidly in such conditions.

If electricity was affordable and reliable, then many microbes could be killed using heat treatment but, in many cities, and especially for the homeless, electricity cannot be relied on.

Inadequate nutrition leads to weaker immune systems, increasing susceptibility to many infectious diseases. Diarrhoeal diseases are caused by the poor sanitation associated with poverty.

Factors that affect the spread of disease

Healthcare provisions

Healthcare provisions include medicine, vaccines, bandages and other medical supplies.

Healthcare services and facilities such as hospitals and available health professionals are also considered to be provisions.

Poor healthcare increases susceptibility to infectious disease. Poor access to medications, particularly preventative medication such as vaccination, leaves many people vulnerable to infectious diseases.

Fewer individuals are able to be immunised against specific diseases. Therefore, there is more risk of a disease spreading and a high proportion of susceptible individuals.